

Overview of Databases

Database Design & Implementation

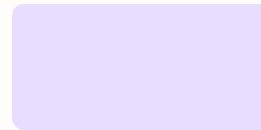
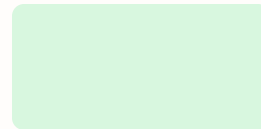
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Overview of Databases

Starting question

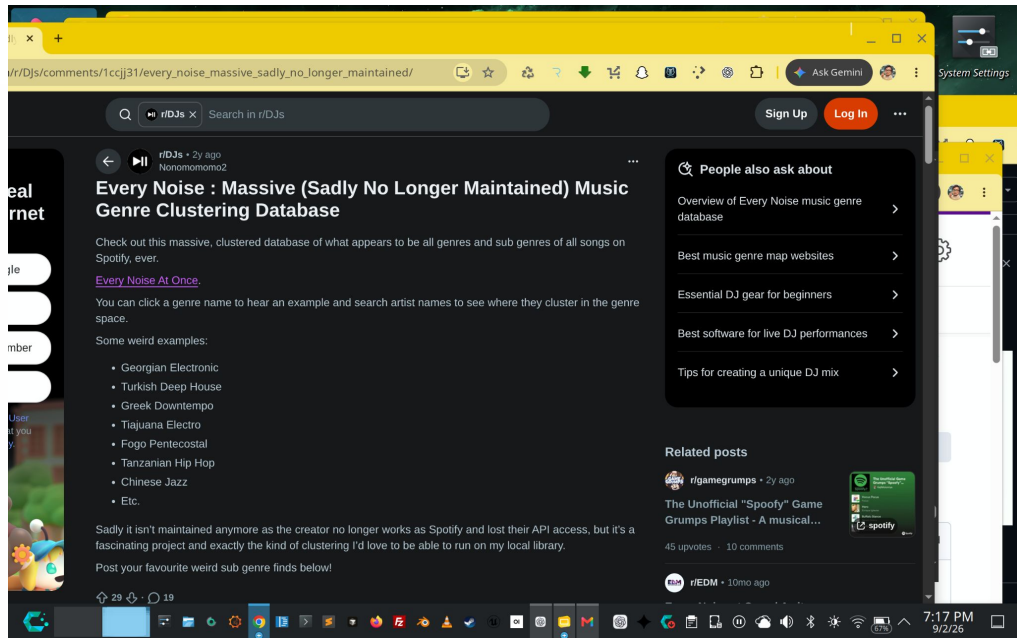
What is a database? Is it different than a file? Is it more specific than record keeping?

I'm interested in your thoughts but first lets look at these





nce scan list playlist **book** more



find artist

latin tech r

re:techno

<https://everynoise.com/>

italian tech r

un tech house

deep tech house

swiss house

deep minimal techr

mexican tech hous

french tech hous

focus trance

german tec

dark techno

brazilian tech

electronica ven

deep progressive house

croatian electronic

house argentino

raw techno

serbian electronic

deep funk house

dark progressive house

spanish techno

italian techno

hard techno

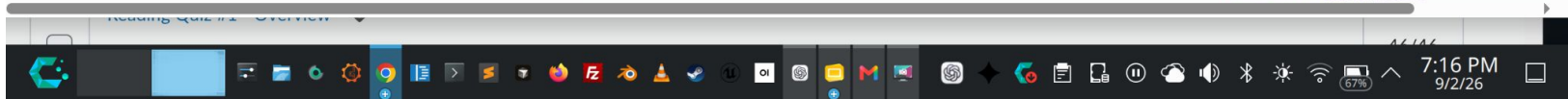
german dark minimal techno

funky tech house

swiss techno

portuguese techno

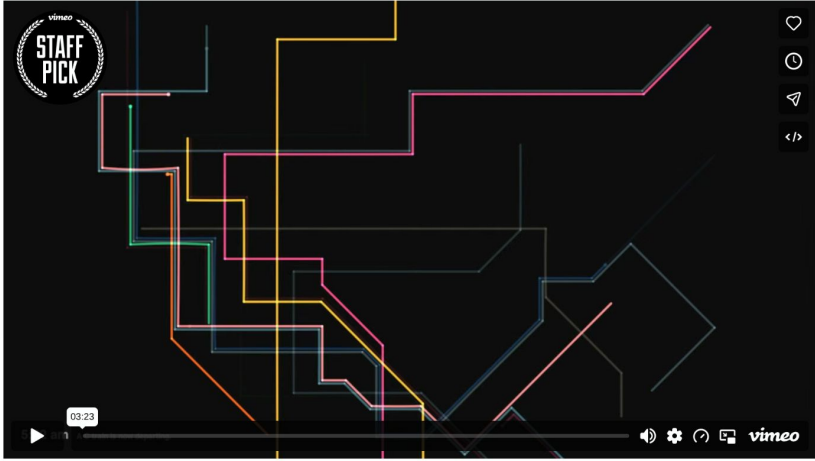
house argentino





<https://www.dear-data.com/theproject>

Mta.me



At www.mta.me, Conductor turns the New York subway system into an interactive string instrument. Using the

In its first week of launch, mta.me was picked up by [The Wall Street Journal](#), [The Huffington Post](#), [Engadget](#), [Gizmodo](#),

<https://www.chenalexander.com/Mta-me>

LLM



Retrieval Augmented Gen



Chromadb



Hugging Face



Code



RAG with HuggingFace models and Chroma db

https://medium.com/@prajwal_/rag-with-huggingface-models-and-chroma-db-3f6ade28b5fe



Prajwal landge

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3 min read · Jul 4, 2024



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What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval and natural language generation. It uses two components:

1. **Retriever:** Searches a large document corpus to find the most relevant

A database organizes data for use

- A database is an organized collection of data.
- The organization matters because people and programs need to find, update, and trust the data.
- Database systems grew out of the limits of program-owned files.



EdStem Demo:

<https://edstem.org/us/courses/105050/workspaces/p8RihV8m9QSoH3rEGGLITUIBa4HEcfAG>

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Workspaces database_lecture_1

constraint_demo.sql

demo_queries.sql

demo.sqlite

erd.png

flat_file_version.csv

README.md

reset_demo.sh

schema.sql

seed.sql

```
11 PRAGMA foreign_keys = ON;
12
13 -- 1. A file can store records. A database lets us ask structured questions.
14 SELECT name, category
15 FROM dishes
16 ORDER BY category, name;
17
18 -- 2. Attributes let us filter precisely.
19 SELECT name
20 FROM dishes
21 WHERE category = 'dessert';
22
23 -- 3. Entities become separate tables.
```

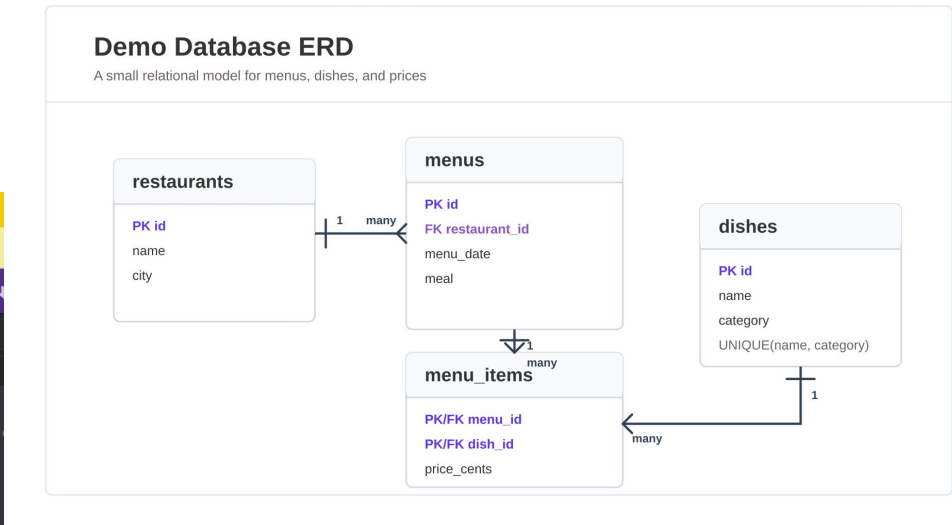
user@sahara:-

ORDER BY city, name;

name	city
Harbor Lunch Room	Boston
The Campus Diner	New York
Union Square Cafe	New York

sqlite>

/home/demo_queries.sql 1:1 Spaces: 4 (Auto)



id	restaurant	city	menu_date	meal	dish	category	price
1	restaurant	New York	2026-09-01	breakfast	Coffee	drink	2.50
2	The Campus ...	New York	2026-09-01	breakfast	Pancakes	breakfast	9.00
3	The Campus ...	New York	2026-09-01	breakfast	Egg Sandwich	breakfast	7.50
4	The Campus ...	New York	2026-09-01	lunch	Coffee	drink	2.50
5	The Campus ...	New York	2026-09-01	lunch	Tomato Soup	soup	6.50
6	The Campus ...	New York	2026-09-01	lunch	Grilled Cheese	sandwich	9.50
7	Union Square ...	New York	2026-09-01	dinner	Roast Chicken	entree	24.00
8	Union Square ...	New York	2026-09-01	dinner	Oysters	seafood	18.00
9	Union Square ...	New York	2026-09-01	dinner	Apple Pie	dessert	8.50
10	Union Square ...	New York	2026-09-02	dinner	Roast Chicken	entree	26.00
11	Union Square ...	New York	2026-09-02	dinner	Oysters	seafood	21.00
12	Union Square ...	New York	2026-09-02	dinner	Apple Pie	dessert	9.00
13	Union Square ...	New York	2026-09-02	dinner	Apple Pie	dessert	9.00

Early databases were closely tied to programs

Early 1960s: databases began as related files

Each file stayed tied to the program using it

Storage details shaped how programs were written

Languages such as COBOL fit these structures



Growing organizations needed more flexibility

The pressure

Computers became more powerful
Business uses expanded quickly
More people needed shared data

The pain

Rigid file structures slowed changes
Teams struggled to coordinate updates
Hardware shifts caused disruption

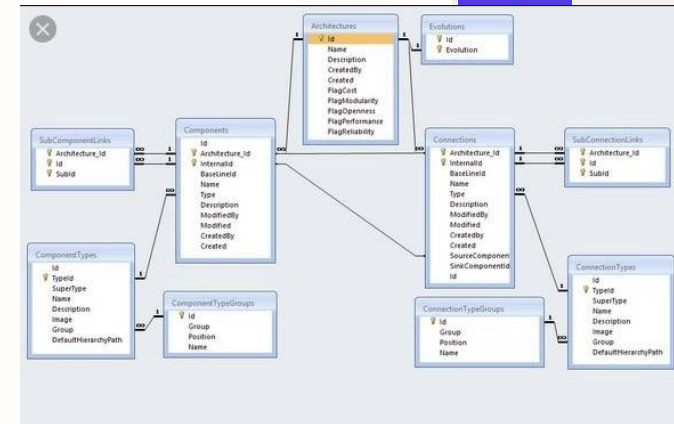
Relational databases won on flexibility

E. F. Codd introduced the relational model in the early 1970s

The model is based on relational mathematics

Adoption took roughly 15 to 20 years

Flexibility outweighed some efficiency tradeoffs



Relational databases changed the model

Separate the views

Application view

How software and users interact with the data.

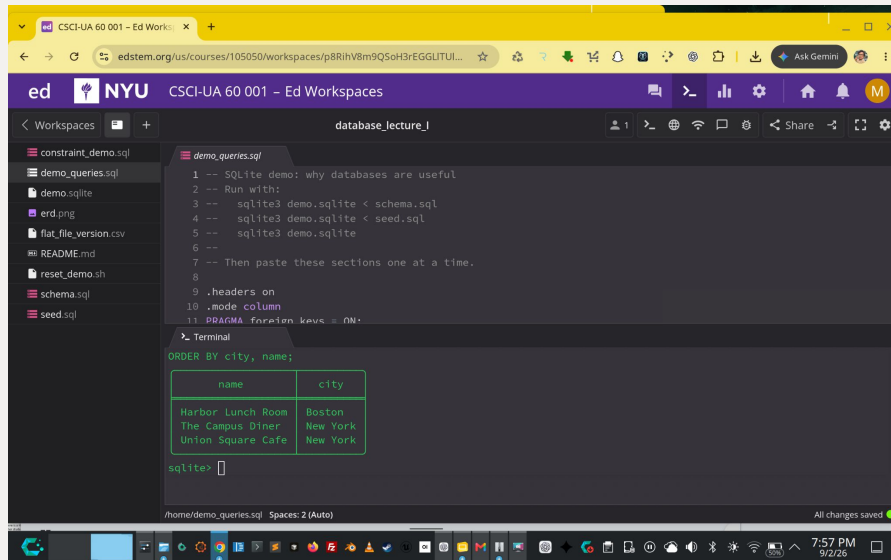
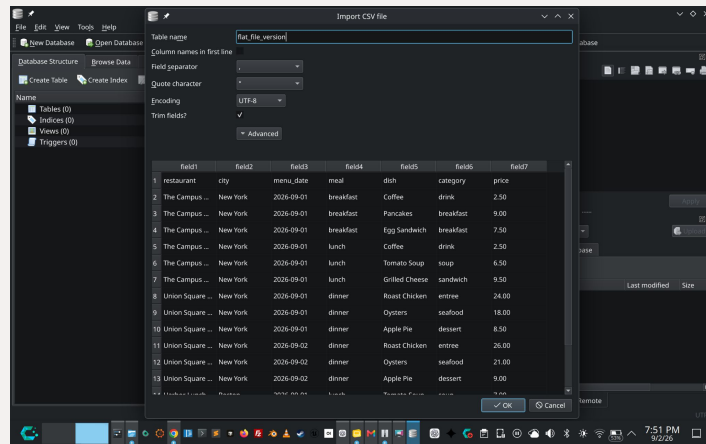
Logical view

How the information is structured conceptually.

Physical view

How data is stored and managed by the system.

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Enterprise systems had to earn trust

Atomicity

All or nothing changes

Consistency

Rules stay valid

Isolation

Work stays separate

Durability

Committed data
persists

Scale

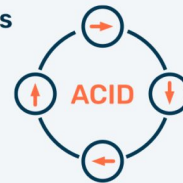
More data and users

Stability

Reliable operations

Yes We Can!

Distributed **ACID** Transactions
with **High Performance**



Relational databases specialized for different jobs

OLTP

Fast day-to-day transactions

Warehouses

Analysis and reporting

In-memory

Speed-sensitive workloads

Lite

Small embedded systems

The landscape now includes SQL and NoSQL choices

Oracle

Enterprise RDBMS

MySQL

Open-source SQL

PostgreSQL

Advanced open source

SQL Server

Microsoft platform

DB2

IBM enterprise

NoSQL

Big, distributed, flexible data



ORACLE®



Introduction to NoSQL

Overview of Databases

-  Stands for Not Only SQL
-  Seeks to solve the need for “Big Data”
-  Supports semi-structured and unstructured data
-  Allows for massively distributed databases
-  Reduces the constraints of relational databases

NoSQL Considerations

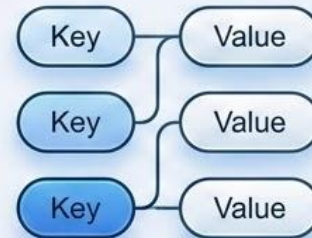
-  Data flexibility over strong consistency tradeoffs
-  Data may be immutable for scale



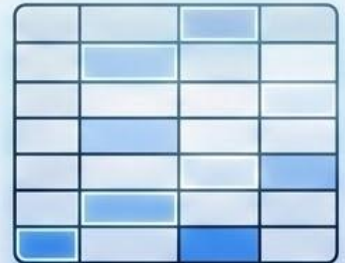
Document



Graph



Key-Value



Wide-column